

Content-Adaptive, Editable Photo Styles: Revising the Project Direction

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Abstract. The project set out to build a “general-purpose stylization engine” in which a fixed network learns any style from data. We argue this goal is ill-posed: its synthetic styles are fixed functions that need no learning. We reframe it as learning the **per-image** parametric edit that takes a photo to a target look, specified by paired or unpaired examples. We then review 18 learned enhancement methods on MIT-Adobe FiveK. Image adaptivity, not spatial processing, carries most of the gain: a fixed 3D LUT reaches 20.37 dB PSNR, Zeng et al.’s image-adaptive blend 25.21 dB, and spatial extensions add at most ≈ 0.5 dB. We recommend a SepLUT-style renderer (per-channel curves, colour matrix, residual basis-LUT blend) driven by a style-conditioned head on frozen features, and a staged evaluation that begins with a data-scaling curve.

1 Introduction

The current system predicts $P = 21$ renderer parameters (a tone curve shared by all channels, a 3×3 colour matrix and bias, grain, vignette) from a 1280-D descriptor: per-channel differentiable CDFs concatenated with ResNet-18 features. It has been trained on three synthetic styles and on FiveK expert C. The synthetic targets are closed-form maps $G(I)$ that apply identical parameters to every image, so the network can at best imitate G , and running G is faster and exact. Only the FiveK run required learning: 80 training pairs reduced held-out L_1 to the expert from 0.0917 to 0.0741 (4/5 images improved), but 500 pairs raised it to 0.1030 (1/5). That result rests on five test images and one seed, and training was confounded: a fully fine-tuned ResNet with BatchNorm at batch 4, no ImageNet normalisation, and unequal epochs.

2 Problem formulation

A **style** s is a target appearance, not an operation. For an input I the model predicts

$$\theta = f(I, s), \quad \hat{I} = R(I; \theta), \quad (1)$$

where R is a differentiable, editable renderer. The style is specified either by **paired** data $\{(I_i, I_i^*)\}$ (a photographer’s edits) or by an **unpaired** set $\mathcal{S}_s$ of in-style photos. The defining property is content adaptivity: θ must vary with I , because a dark street and a bright beach need different edits to reach the same look. Constraints: CPU training, 10^1 – 10^2 examples per style, outputs exportable as curves or a .cube LUT, no generated content.

3 Background: image-adaptive 3D LUTs

Zeng et al. [1] learn $N = 3$ basis LUTs Φ_n on a 33^3 RGB lattice. A 269K-parameter CNN sees a 256^2 thumbnail and predicts weights $w_n(I)$; the image’s LUT is

$$\Phi(I) = \sum_{n=1}^N w_n(I) \Phi_n, \quad (2)$$

applied at full resolution by trilinear interpolation. Training minimises $\mathcal{L}_{\text{MSE}} + 10^{-4} \mathcal{R}_s + 10 \mathcal{R}_m$, where $\mathcal{R}_s$ is a total-variation smoothness term and $\mathcal{R}_m$ penalises non-monotone outputs. The model has 593.5K parameters and runs in 1.66 ms per 4K image. On FiveK (expert C, 480p) it scores 25.21 dB paired and 22.86 dB when trained unpaired with a WGAN-GP critic. Its ablation is the key evidence for this project: one **fixed** LUT scores 20.37 dB, one adaptive LUT 23.15 dB, three 25.21 dB.

4 Comparative analysis

Table 1 lists representative methods. Accuracy has converged within ≈ 0.5 dB, so size, conditioning and editability decide.

Method	dB	Params	MS	UP	Ed
3D LUT [1]	25.21	593K	–	✓	LUT
AdaInt [2]	25.49	620K	–	–	–
SepLUT-S [3]	25.42	47K	–	–	curves+LUT
SA-3DLUT [4]	25.50	4.5M	–	–	–
CSRNet [5]	25.17	36K	–	–	–
HDRNet [6]	24.66	482K	–	–	–
StarEnhancer [7]	25.29†	n/r	✓	few-shot	curves
PieNet [8]	25.28†	n/r	✓	few-shot	–
Exposure [9]	21.32	8.6M	–	✓	filters
InstructP2P [10]	n/a	>860M	text	–	–

Table 1: FiveK expert C PSNR (480p; † other protocol). MS: multi-style; UP: unpaired; Ed: editable output; n/r: not reported.

Four findings follow. (i) **Adaptivity dominates**: spatially aware LUTs add 0.26 dB on FiveK [4], and global CSRNet beats local HDRNet. (ii) **Separable is cheaper**: per-channel 1D curves before a 9^3 LUT match Zeng at one-twelfth the size, and SepLUT’s LUT generator is Zeng’s basis blend [3]. (iii) **Only StarEnhancer** combines multiple styles, few-shot new styles from example images, and editable curves [7]; CSRNet shows fine-tuning only the conditioning matches retraining for a new expert [5]. (iv) **Unpaired training is costly**: -2.35 dB for Zeng, and most unpaired rivals barely beat Camera Raw’s auto setting (21.61 dB) [1]. Pixel generators and diffusion editors can alter content [10] and are excluded.

5 Proposed approach

Features. Keep ResNet-18 + CDF, frozen and cached, with ImageNet normalisation. **Renderer.** A cascade of three per-channel monotone curves, the 3×3 matrix and bias, and a residual 9^3 LUT $\sum_n w_n \Phi_n$ over $N \approx 4$ –8 shared bases with Zeng’s regularisers, followed by grain and vignette. The composite bakes into one 33^3 .cube. **Conditioning.** A small head without BatchNorm, modulated by FiLM on a style code: a learned embedding for known styles, or the mean embedding of n example images for new ones [7], [8]. **Unpaired styles.** A batch-level CDF / sliced-Wasserstein colour loss, an identity anchor and distort-and-recover pseudo-pairs, with a small WGAN-GP critic only if needed.

6 Limitations and evaluation plan

No study reports these methods below 675 training pairs; the combination is untested; style codes from few images of subtle photographers are noisy (StarEnhancer’s expert Recall@1 is as low as 24.6%); the LUT stage is not slider-editable. We therefore gate each step: (1) remove the confounds and measure a learning curve from 25 to 4,500 pairs (3 seeds, ≥ 100 test images, identity, mean-edit, histogram-matching and per-image oracle baselines); (2) ablate renderers against CSRNet; (3) train on experts A–D and hold out E; (4) test unpaired losses on disjoint FiveK images; (5) verify .cube export.

7 Conclusion

The project’s value lies where no closed-form edit exists: content-dependent looks learned from examples. The evidence favours a compact, global, image-adaptive and style-conditioned renderer over larger spatial or generative models. The first deliverable is the learning curve, not a new model.

7 References

- [1] H. Zeng, J. Cai, L. Li, Z. Cao, and L. Zhang, “Learning Image-adaptive 3D Lookup Tables for High Performance Photo Enhancement in Real-time,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020, [Online]. Available: <https://arxiv.org/abs/2009.14468>
- [2] C. Yang, M. Jin, X. Jia, Y. Xu, and Y. Chen, “AdaInt: Learning Adaptive Intervals for 3D Lookup Tables on Real-time Image Enhancement,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. [Online]. Available: <https://arxiv.org/abs/2204.13983>
- [3] C. Yang, M. Jin, Y. Xu, R. Zhang, Y. Chen, and H. Liu, “SepLUT: Separable Image-adaptive Lookup Tables for Real-time Image Enhancement,” in *European Conference on Computer Vision (ECCV)*, 2022. [Online]. Available: <https://arxiv.org/abs/2207.08351>
- [4] T. Wang *et al.*, “Real-time Image Enhancer via Learnable Spatial-aware 3D Lookup Tables,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/papers/Wang_Real-Time_Image_Enhancer_via_Learnable_Spatial-Aware_3D_Lookup_Tables_ICCV_2021_paper.pdf
- [5] J. He, Y. Liu, Y. Qiao, and C. Dong, “Conditional Sequential Modulation for Efficient Global Image Retouching,” in *European Conference on Computer Vision (ECCV)*, 2020. [Online]. Available: <https://arxiv.org/abs/2009.10390>
- [6] M. Gharbi, J. Chen, J. T. Barron, S. W. Hasinoff, and F. Durand, “Deep Bilateral Learning for Real-Time Image Enhancement,” *ACM Transactions on Graphics (SIGGRAPH)*, 2017, [Online]. Available: <https://arxiv.org/abs/1707.02880>
- [7] Y. Song, H. Qian, and X. Du, “StarEnhancer: Learning Real-Time and Style-Aware Image Enhancement,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2021/papers/Song_StarEnhancer_Learning_Real-Time_and_Style-Aware_Image_Enhancement_ICCV_2021_paper.pdf
- [8] H.-U. Kim, Y. J. Koh, and C.-S. Kim, “PieNet: Personalized Image Enhancement Network,” in *European Conference on Computer Vision (ECCV)*, 2020. [Online]. Available: https://www.ecva.net/papers/eccv_2020/papers_ECCV/papers/123750375.pdf
- [9] Y. Hu, H. He, C. Xu, B. Wang, and S. Lin, “Exposure: A White-Box Photo Post-Processing Framework,” *ACM Transactions on Graphics*, 2018, [Online]. Available: <https://arxiv.org/abs/1709.09602>
- [10] T. Brooks, A. Holynski, and A. A. Efros, “InstructPix2Pix: Learning to Follow Image Editing Instructions,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. [Online]. Available: <https://arxiv.org/abs/2211.09800>